

Building a CI/CD Workflow



Ned Bellavance

HashiCorp Ambassador

@ned1313 | nedinthecloud.com



Module Overview



Use versioned source control

Add CI/CD automation

- Continuous integration
- Continuous delivery

DevOps primer



Globomantics Environment



GLOBOMANTICS

Move Terraform code to GitHub

Add CI with GitHub Actions

Add CD with Terraform Cloud

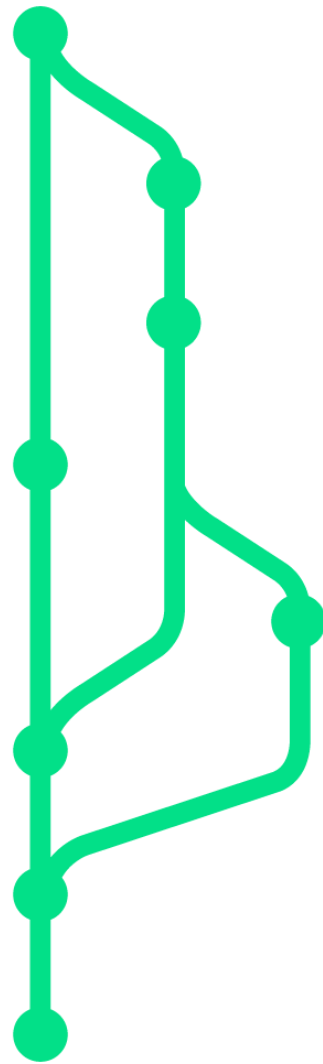




Terminology for DevOps



Source Control Management



Multiple formats

- Git, TFVC, Subversion

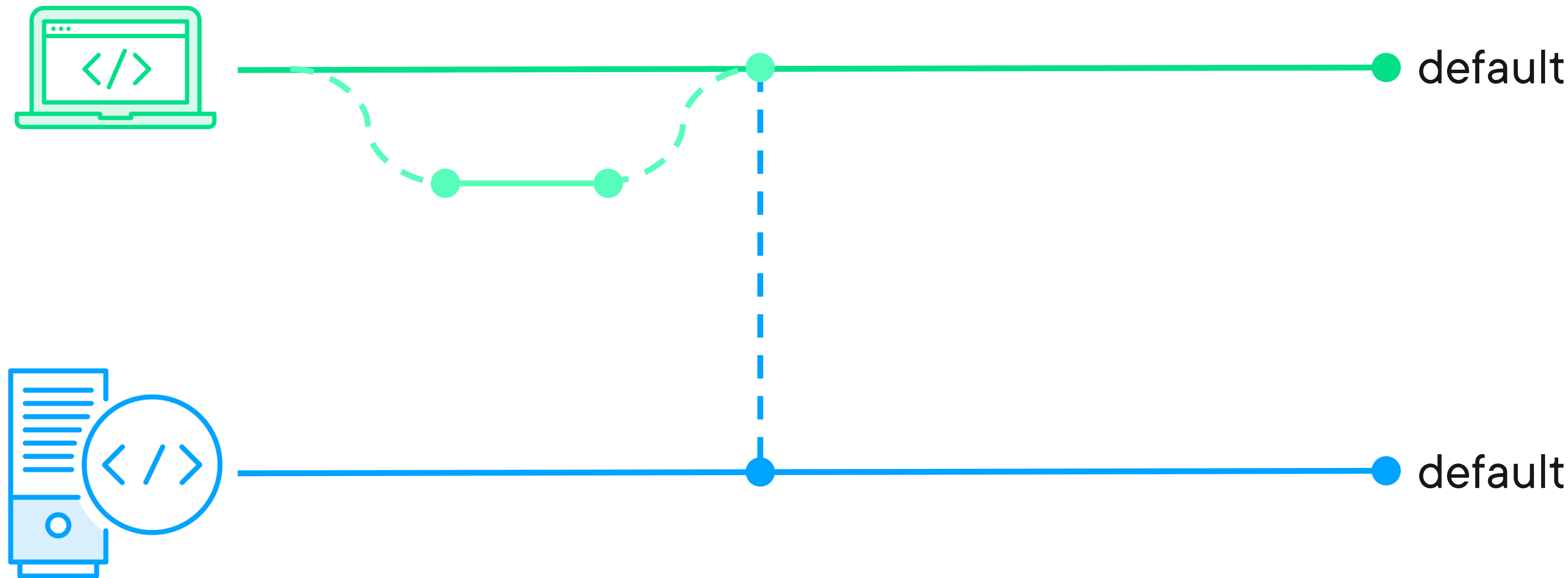
Multiple platforms

- GitHub, BitBucket, GitLab, CodeCommit

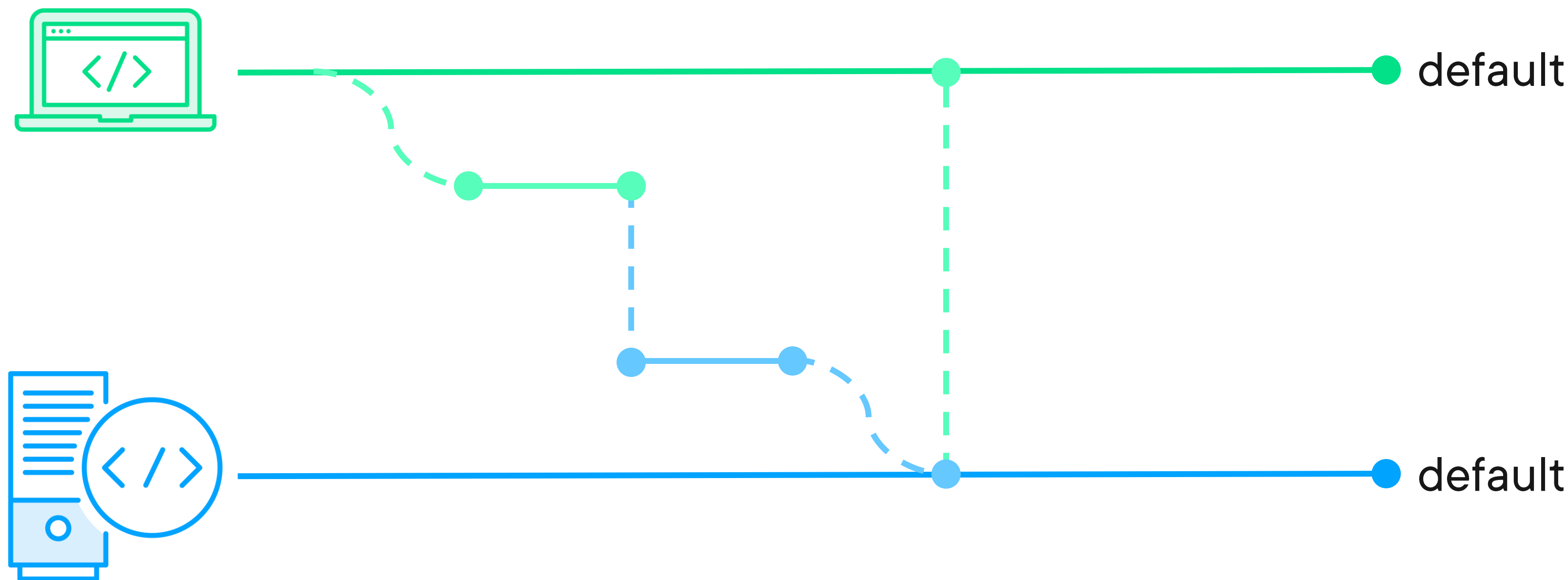
Enable collaboration



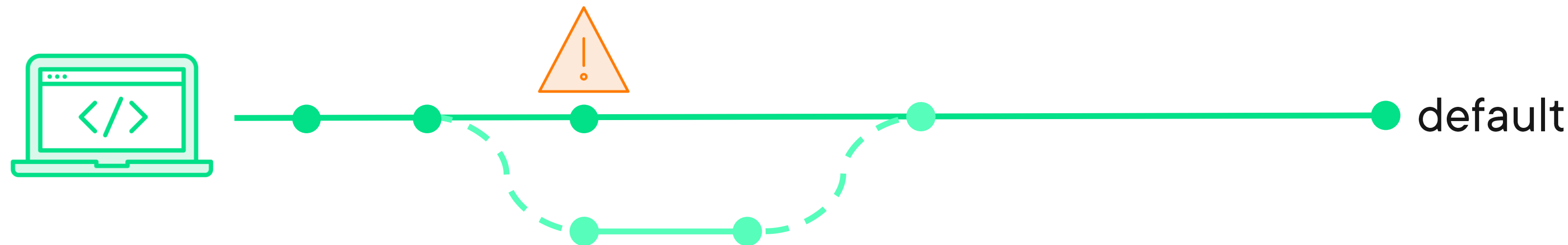
Git and GitHub



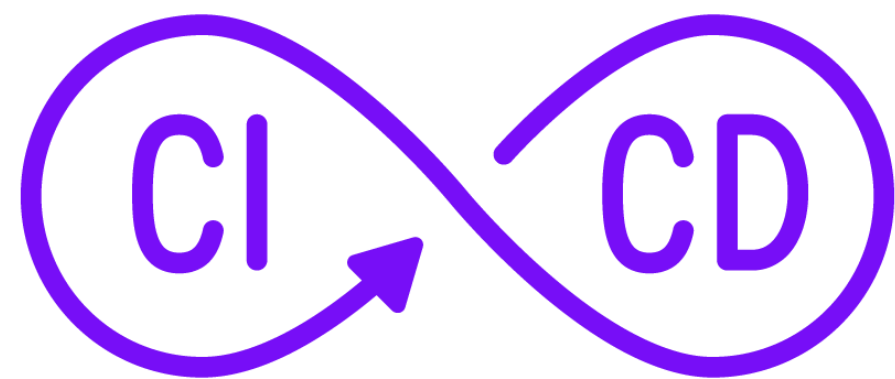
Git and GitHub



Version Control



Continuous Integration/Continuous Delivery



Multiple platforms

- Jenkins, CodePipeline, Azure DevOps

Continuous integration

- Check validity and policy
- Test integration with existing code

Continuous delivery

- Deliver build artifacts
- Automated testing and validation
- Temporary environments

Multiple environments





Source control with Git and GitHub

Prerequisites

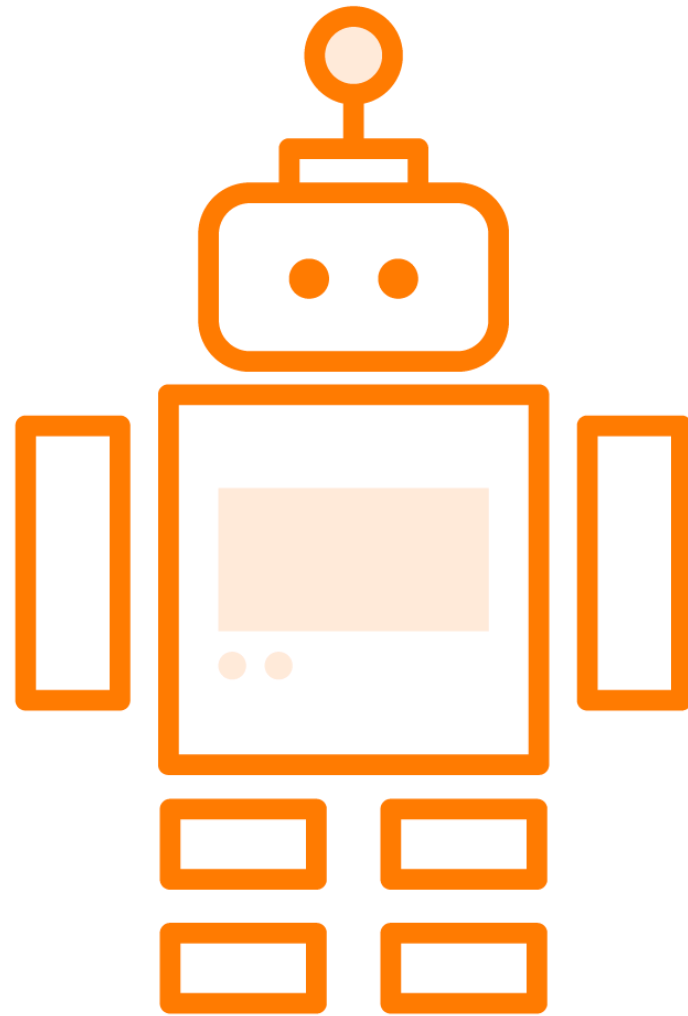
- Git CLI installed
- GitHub account

Process

- Initialize code with Git
- Create GitHub repository
- Push code to GitHub



Automation Considerations



Files to exclude from source control

- .gitignore

Provider lock file

- .terraform.lock.hcl

Plugin caching

- TF_PLUGIN_CACHE_DIR

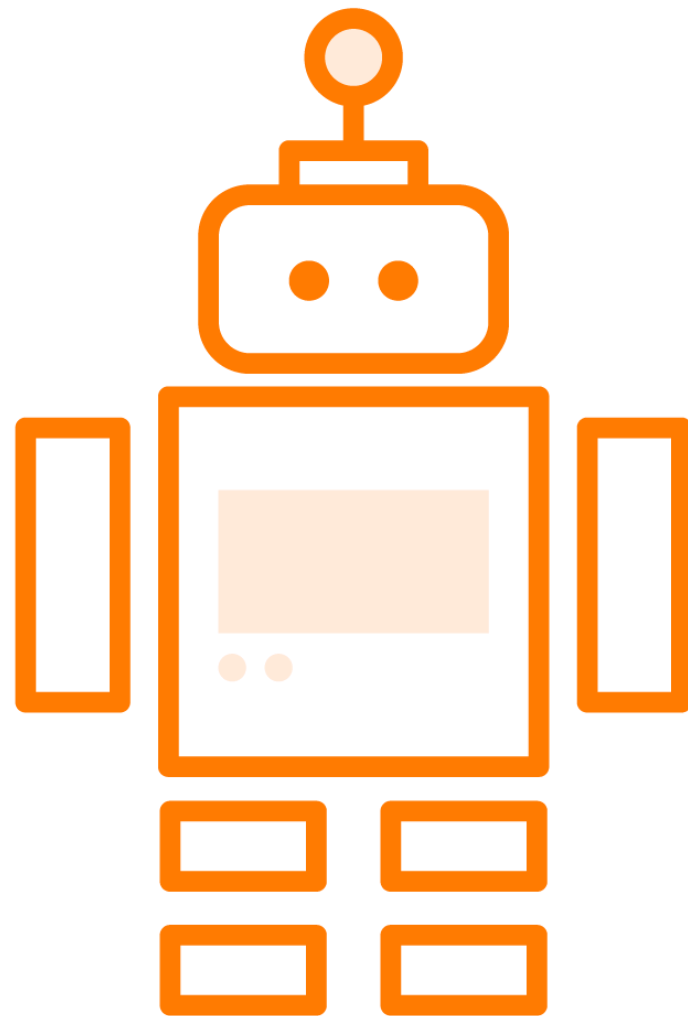
Backend and provider credentials

Terminal output control

- TF_IN_AUTOMATION



Automation Considerations



Deployment process

- Save execution plan?
- Enable continuous deployment?

Error handling

- Continue or cancel
- Log storage
- Log level



Automation Environment Variables

TF_IN_AUTOMATION = TRUE

TF_LOG = "INFO"

TF_LOG_PROVIDER = "ERROR"

TF_LOG_PATH = "FILE_PATH"

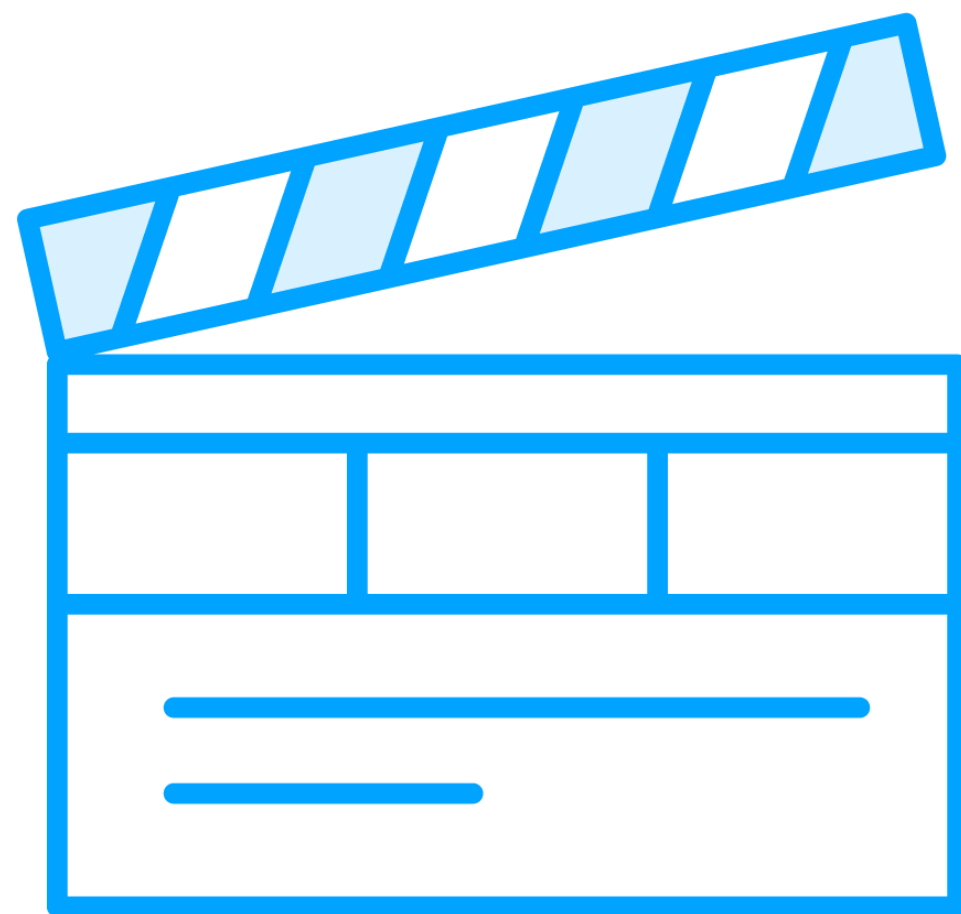
TF_INPUT = FALSE

TF_VAR_name = "VALUE"

TF_CLI_ARGS = "COMMAND_FLAGS"



GitHub Actions



Triggered on event or schedule

Included with repository

Library of actions available

What to check?

- Formatting and validity

Need to initialize Terraform





Terraform Cloud Integration



Globomantics Environment



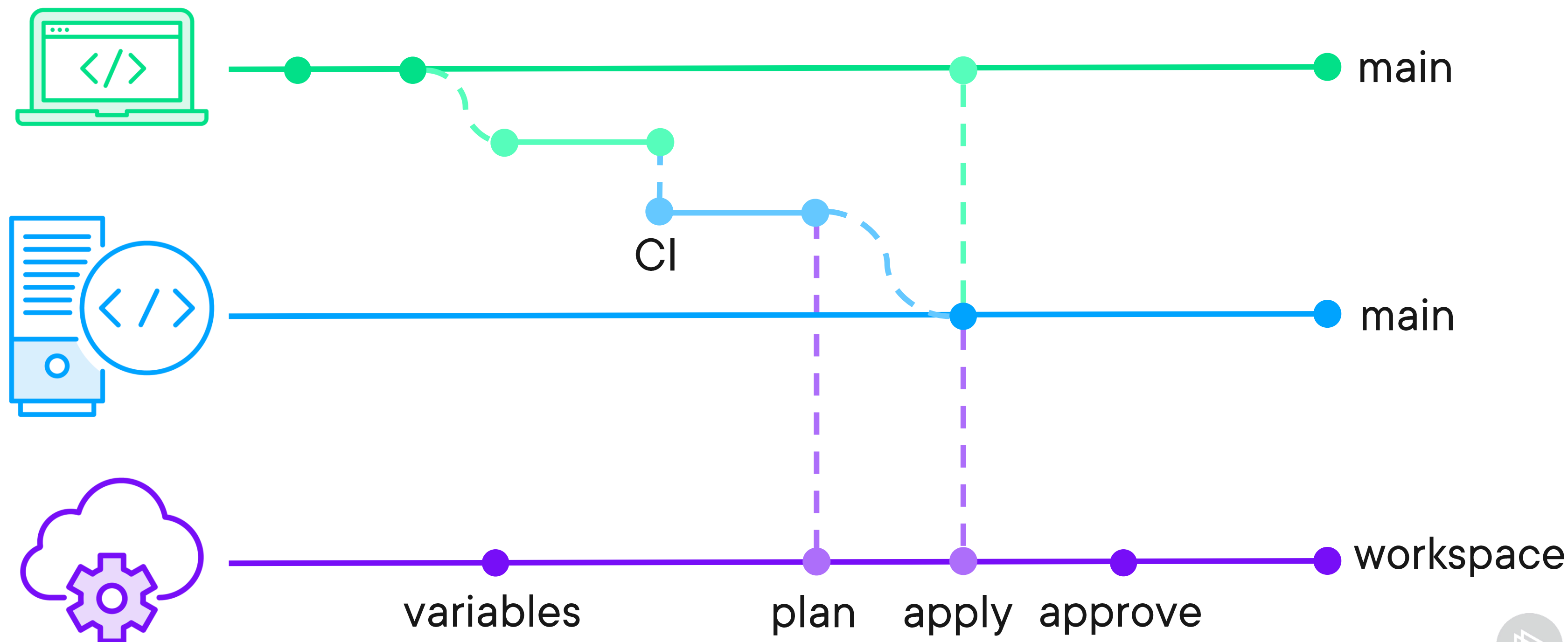
GLOBOMANTICS

Requested changes

- Add a third public subnet
- Add tag for billing code



CI/CD Workflow



Module Summary



Protect and share your code

Automate testing and validation

Trigger on VCS events



Up Next:

Supporting Multiple Environments

